

Def. Let V be a vector space.

Given a vector $\alpha \in V$ and linear operator $T: V \rightarrow V$, the subspace

$$Z(\alpha, T) \stackrel{\text{def}}{=} \{g(T)\alpha \mid g(t) \in F[t]\} = \text{span}\{T^k \alpha \mid k \in \mathbb{N} \cup \{0\}\}.$$

is called the T -cyclic subspace generated by α

Def. Let V be a vector space. Given $\alpha \in V$ and a linear operator T on V , an ideal

$$M(\alpha, T) = \{g(t) \in F[t] \mid g(T)\alpha = 0\}$$

is called the T -annihilator of α , and the unique monic generator is also called ".

Thm. Let α be any non-zero vector in V and let p_α be the T -annihilator of α .

- 1) $\deg p_\alpha = \dim Z(\alpha, T)$
- 2) $Z(\alpha, T) = \text{span}\{T^k \alpha \mid 0 \leq k \leq \deg p_\alpha\}$
- 3) The minimal polynomial of $T|_{Z(\alpha, T)}$ is p_α .

Proof. Given a polynomial $g \in F[x]$, the Euclidean Algorithm gives

$$g = p_\alpha \cdot q + r \quad \text{for some } q, r \in F[x], \text{ with } r = 0 \text{ or } \deg r < \deg p_\alpha.$$

Then, $g(T)\alpha = p_\alpha(T)q(T)\alpha + r(T)\alpha = r(T)\alpha$. Since either $r = 0$ or $\deg r < \deg p_\alpha$, the vector $r(T)\alpha$ is a linear combination of $\{\alpha, T\alpha, T^2\alpha, \dots, T^{\deg p_\alpha - 1}\alpha\}$.

Since g was arbitrary, $Z(\alpha, T)$ is spanned by $\{T^n \alpha \mid n = 0, 1, \dots, \deg p_\alpha - 1\}$

Moreover, these are linearly independent, because: if there exists non-trivial representation

$$c_0 \alpha + c_1 T\alpha + \dots + c_{\deg p_\alpha - 1} T^{\deg p_\alpha - 1} \alpha = 0,$$

then it contradicts to the minimality of p_α . It proves 1) and 2).

To show the third assertion, let $U = T|_{Z(\alpha, T)}: Z(\alpha, T) \rightarrow Z(\alpha, T): \alpha \mapsto T\alpha$.

It is well-defined, since the cyclic subspace is invariant under T .

Given a polynomial $g \in F[x]$,

$$p_\alpha(U)g(T)\alpha \underset{2)}{=} p_\alpha(T)g(T)\alpha = g(T)p_\alpha(T)\alpha = g(T)0 = 0.$$

$$g(T)\alpha \in Z(\alpha, T)$$

So, $p_\alpha(U)$ is identically zero. Moreover, $h(U)$ cannot be zero if $\deg h < \deg p_\alpha$, because $h(U)\alpha = h(T)\alpha = 0$, then it contradicts to the minimality.

Companion Matrix.

Let W be a vector space of dimension $K \in \mathbb{N}$, and $T: W \rightarrow W$ be a linear operator which have a cyclic vector $\alpha \in W$. By the Theorem above, $\{\alpha, T\alpha, \dots, T^{K-1}\alpha\}$ forms a basis for W . As above fact and the Cayley-Hamilton Theorem, let

$$P_\alpha(x) = x^K + C_{K-1}x^{K-1} + \dots + C_1x + C_0$$

be the minimal polynomial (and also characteristic) for T .

Define $\beta = \{T^{n-1}\alpha \mid n=1, 2, \dots, K\}$, denote $d_n = T^{n-1}\alpha$. Then,

$$0 = 0 \cdot \alpha = P_\alpha(T)\alpha = T^K\alpha + C_{K-1}T^{K-1}\alpha + \dots + C_1T\alpha + C_0\alpha$$

$$\Rightarrow T^K\alpha = -C_0\alpha - C_1T\alpha - \dots - C_{K-1}T^{K-1}\alpha$$

$$= -C_0d_1 - C_1d_2 - \dots - C_{K-1}d_K$$

$$[T]_\beta = \begin{bmatrix} 0 & 0 & \dots & 0 & -C_0 \\ 1 & 0 & \dots & 0 & -C_1 \\ 0 & 1 & \dots & 0 & -C_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & -C_{K-1} \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 & -C_0 \\ & \ddots & & \vdots \\ & & I_{K-1} & \vdots \\ & & & -C_{K-1} \end{bmatrix}$$

Directly, we obtain that: on a fin. dim space T ,

A linear operator T has a cyclic vector

if and only if

the minimal polynomial of T and characteristic polynomial of T are identical.

Exercises in Hoffman, Sec 7.2

1. Every linear operator T on two-dimensional either:

T has a cyclic vector or T is a scalar multiple of the identity.

Proof. Suppose that T has no cyclic vector. That is,